CLASSWORK 11

Logistics Regression by Hand

CPE 486/586

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20 points

A maintenance team monitors cooling fans in a server room. For a fan, they record two integer readings on a four-level scale with values 0, 1, 2 and 3. Here x_1 is the vibration level and x_2 is the lubrication level. The label is $y = 1$ when the fan failed within 30 days and $y = 0$ when it kept running. Table 1 lists the five fans used for training.

Table 1. Training data, $N = 5$.

Fan i	x_{i1} (vibration)	x_{i2} (lubrication)	y_i
1	3	0	1
2	1	2	0
3	3	1	1
4	0	1	0
5	2	2	0

A new technician has some initial weight to start with. You'll start from these weights and work on determining how gradient descent evolve them.

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \\ 0.5 \end{bmatrix}, \qquad \eta = 1.0, \qquad N = 5.$$

Equations to use

Augmented input:	$\mathbf{x}^{(i)} = \begin{bmatrix} 1 & x_{i1} & x_{i2} \end{bmatrix}^\top$, the leading 1 for the bias θ_0
Linear score:	$z_i = \boldsymbol{\theta}^\top \mathbf{x}^{(i)} = \theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2}$
Hypothesis:	$h_i = h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) = g(z_i) = \dfrac{1}{1+e^{-z_i}}$
Cost (natural log):	$J(\boldsymbol{\theta}) = -\dfrac{1}{N}\sum_{i=1}^{N}\Big[y_i \ln h_i + (1-y_i)\ln(1-h_i)\Big] = \dfrac{1}{N}\sum_{i=1}^{N}\ell_i$
Per-sample loss:	$\ell_i = -\ln h_i$ when $y_i = 1$, $\quad \ell_i = -\ln(1-h_i)$ when $y_i = 0$
Error:	$e_i = h_i - y_i$
Gradient:	$\dfrac{\partial J}{\partial \theta_j} = \dfrac{1}{N}\sum_{i=1}^{N} e_i\, x_{ij} = \dfrac{S_j}{N}$, $\quad$ with $x_{i0} = 1$
Update:	$\theta_j^{(t+1)} = \theta_j^{(t)} - \eta \left.\dfrac{\partial J}{\partial \theta_j}\right\vert_{\boldsymbol{\theta}^{(t)}}$ $\quad$ for $j = 0, 1, 2$
Prediction:	$\hat{y} = 1$ when $h \geq 0.5$, which happens exactly when $z \geq 0$
Decision boundary:	$\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$, $\quad$ so $\quad x_2 = -\dfrac{\theta_0 + \theta_1 x_1}{\theta_2}$

Sigmoid Properties

(P1)	$g(0) = \frac{1}{2}$	A score of zero at the boundary.
(P2)	$g(-z) = 1 - g(z)$	
(P3)	$\ln\big(1 - g(z)\big) = \ln g(-z)$	For $z < 0$, the two loss columns of Table 2 swap.
(P4)	$g'(z) = g(z)\big(1 - g(z)\big)$	With the chain rule, it gives $\partial \ell_i / \partial z_i = h_i - y_i$.

A note on signs. The slides define the log-likelihood $LL(\boldsymbol{\theta}) = \sum_i \big[y_i \ln h_i + (1-y_i)\ln(1-h_i)\big]$, so $LL = -NJ$. Maximizing LL and minimizing J lead to the same $\boldsymbol{\theta}$. Here you minimize J, so the update subtracts the gradient of J. In the form of the slides, $\theta_j \leftarrow \theta_j + \frac{\eta}{N}\sum_i (y_i - h_i)\, x_{ij}$, which is the same step.

Rules for Hand Computation

R1. Compute z_i with max three decimal places.

R2. Round z_i to the nearest 0.1 only to read Table 2. Round a halfway value away from zero.

R3. Copy h_i and ℓ_i from Table 2 with four decimals. Keep four decimals for e_i, $e_i x_{ij}$, S_j and $\partial J / \partial \theta_j$.

R4. Round $\theta_j^{(t+1)}$ to three decimals before the next iteration.

R5. Report J with four decimals.

Steps

1. Compute $z_i = \theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2}$ with the current $\boldsymbol{\theta}^{(t)}$.
2. Read $h_i = g(z_i)$ from Table 2.
3. Read ℓ_i from Table 2, in the column that matches y_i.
4. Compute $e_i = h_i - y_i$.
5. Compute $e_i x_{i1}$ and $e_i x_{i2}$.
6. Add down the columns. $S_0 = \sum_i e_i$, $S_1 = \sum_i e_i x_{i1}$, $S_2 = \sum_i e_i x_{i2}$.
7. Set $\partial J/\partial \theta_j = S_j/5$ and $\theta_j^{(t+1)} = \theta_j^{(t)} - \eta\, \partial J/\partial \theta_j$. Update $j = 0, 1, 2$ from the old $\boldsymbol{\theta}^{(t)}$ together.
8. Compute $J(\boldsymbol{\theta}^{(t)}) = \frac{1}{5}\sum_i \ell_i$ and count the fans with $\hat{y}_i = y_i$. These describe the weights before the update.

Table 2. Sigmoid values and per-sample losses for $z \geq 0$. Apply (P2) and (P3) for $z < 0$.

z	$g(z)$	$-\ln g(z)$ loss if $y = 1$	$-\ln(1 - g(z))$ loss if $y = 0$
0.0	0.5000	0.6931	0.6931
0.1	0.5250	0.6444	0.7444
0.2	0.5498	0.5981	0.7981
0.3	0.5744	0.5544	0.8544
0.4	0.5987	0.5130	0.9130
0.5	0.6225	0.4741	0.9741
0.6	0.6457	0.4375	1.0375
0.7	0.6682	0.4032	1.1032
0.8	0.6900	0.3711	1.1711
0.9	0.7109	0.3412	1.2412
1.0	0.7311	0.3133	1.3133
1.1	0.7503	0.2873	1.3873
1.2	0.7685	0.2633	1.4633
1.3	0.7858	0.2410	1.5410
1.4	0.8022	0.2204	1.6204
1.5	0.8176	0.2014	1.7014
1.6	0.8320	0.1839	1.7839
1.7	0.8455	0.1678	1.8678
1.8	0.8581	0.1530	1.9530
1.9	0.8699	0.1394	2.0394
2.0	0.8808	0.1269	2.1269
2.1	0.8909	0.1155	2.2155
2.2	0.9002	0.1051	2.3051
2.3	0.9089	0.0955	2.3955
2.4	0.9168	0.0868	2.4868
2.5	0.9241	0.0789	2.5789
2.6	0.9309	0.0716	2.6716
2.7	0.9370	0.0650	2.7650
2.8	0.9427	0.0590	2.8590
2.9	0.9478	0.0536	2.9536
3.0	0.9526	0.0486	3.0486

Example of a negative lookup. Take $z = -0.2$. Property (P2) gives $g(-0.2) = 1 - 0.5498 = 0.4502$. For $y = 1$, the loss is $-\ln 0.4502$, read from the $y = 0$ column at 0.2, so $\ell = 0.7981$. For $y = 0$, the loss comes from the $y = 1$ column at 0.2, so $\ell = 0.5981$.

Part 1: Iteration 1, $\boldsymbol{\theta}^{(0)} \to \boldsymbol{\theta}^{(1)}$ (5 points)

Start from $\boldsymbol{\theta}^{(0)} = (\theta_0, \theta_1, \theta_2) = (0.5,\ -0.5,\ 0.5)$.

(a) **[3 points]** Complete Table 3 with the first six steps.

Table 3. Forward pass and errors at $\boldsymbol{\theta}^{(0)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$e_i = h_i - y_i$	$e_i\, x_{i1}$	$e_i\, x_{i2}$
1	3	0	1	-1.0	0.2689	1.3133	-0.7311	-2.1933	0
2	1	2	0	1.0	0.7311	1.3133	0.7311	0.7311	1.4622
3	3	1	1	-0.5	0.3775	0.9741	-0.6225	-1.8675	-0.6225
4	0	1	0	1.0	0.7311	1.3133	0.7311	0	0.7311
5	2	2	0	0.5	0.6225	0.9741	0.6225	1.2450	1.2450
					Column sums	5.8881	$S_0 = 0.7311$	$S_1 = -2.0847$	$S_2 = 2.8158$

(b) **[2 points]** Complete Table 4 with step 7, then finish step 8.

Table 4. Gradient and update, $\eta = 1.0$.

j	$\theta_j^{(0)}$	S_j (Table 3)	$\frac{\partial J}{\partial \theta_j} = \frac{S_j}{5}$	$\theta_j^{(1)} = \theta_j^{(0)} - \eta \frac{\partial J}{\partial \theta_j}$
0	0.5	0.7311	0.1462	0.354
1	-0.5	-2.0847	-0.4169	-0.083
2	0.5	2.8158	0.5632	-0.063

$J(\boldsymbol{\theta}^{(0)}) = \frac{1}{5}\sum_i \ell_i =$ $5.8881/5 = 1.1776$ Fans with $\hat{y}_i = y_i$ under $\boldsymbol{\theta}^{(0)}$: 0 of 5

Solution

Before the update, $\boldsymbol{\theta}^{(0)}$ gets the five fans wrong. Fans 1 and 3 failed, and their negative scores predict that they keep running. Fans 2, 4 and 5 kept running, and their positive scores predict failure. The smallest per-sample loss is 0.9741, well above $\ln 2 = 0.6931$, the loss of a model that answers 0.5.

The update raises θ_1 and lowers θ_2, toward the signs the data support. θ_2 crosses zero and lands at -0.063. θ_1 moves to -0.083. The new $\boldsymbol{\theta}^{(1)}$ sits close to the origin.

Part 2: Iteration 2, $\boldsymbol{\theta}^{(1)} \to \boldsymbol{\theta}^{(2)}$ (5 points)

Copy $\boldsymbol{\theta}^{(1)}$ from Table 4. $\theta_0 =$ 0.354 $\theta_1 =$ −0.083 $\theta_2 =$ −0.063

(a) **[3 points]** Complete Table 5 with the first six steps.

Table 5. Forward pass and errors at $\boldsymbol{\theta}^{(1)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$e_i = h_i - y_i$	$e_i\, x_{i1}$	$e_i\, x_{i2}$
1	3	0	1	$0.105 \to 0.1$	0.5250	0.6444	−0.4750	−1.4250	0
2	1	2	0	$0.145 \to 0.1$	0.5250	0.7444	0.5250	0.5250	1.0500
3	3	1	1	$0.042 \to 0.0$	0.5000	0.6931	−0.5000	−1.5000	−0.5000
4	0	1	0	$0.291 \to 0.3$	0.5744	0.8544	0.5744	0	0.5744
5	2	2	0	$0.062 \to 0.1$	0.5250	0.7444	0.5250	1.0500	1.0500
				Column sums		3.6807	$S_0 = 0.6494$	$S_1 = -1.3500$	$S_2 = 2.1744$

(b) **[2 points]** Complete Table 6 with step 7, then finish step 8.

Table 6. Gradient and update, $\eta = 1.0$.

j	$\theta_j^{(1)}$	S_j (Table 5)	$\frac{\partial J}{\partial \theta_j} = \frac{S_j}{5}$	$\theta_j^{(2)} = \theta_j^{(1)} - \eta \frac{\partial J}{\partial \theta_j}$
0	0.354	0.6494	0.1299	0.224
1	−0.083	−1.3500	−0.2700	0.187
2	−0.063	2.1744	0.4349	−0.498

$J(\boldsymbol{\theta}^{(1)}) = \frac{1}{5}\sum_i \ell_i =$ 3.6807/5 = 0.7361 Fans with $\hat{y}_i = y_i$ under $\boldsymbol{\theta}^{(1)}$: 2 of 5

Solution

With $\boldsymbol{\theta}^{(1)}$ near zero, the five values of h_i stay close to 0.5, the largest being 0.5744, and $J(\boldsymbol{\theta}^{(1)}) = 0.7361$ sits near $\ln 2$. The scores are positive, so the model predicts failure for the five fans and gets 2 right. Fan 3 has $z_3 = 0.042$, which rounds to 0 for the lookup. The sign of the unrounded score decides the prediction, so $\hat{y}_3 = 1$.

After this update, $\theta_1 = 0.187$ is positive and $\theta_2 = -0.498$ is negative. The orientation of $\boldsymbol{\theta}$ has flipped.

Part 3: Iteration 3, $\boldsymbol{\theta}^{(2)} \to \boldsymbol{\theta}^{(3)}$ (5 points)

Copy $\boldsymbol{\theta}^{(2)}$ from Table 6. $\theta_0 =$ 0.224 $\theta_1 =$ 0.187 $\theta_2 =$ −0.498

(a) **[3 points]** Complete Table 7 with the first six steps.

Table 7. Forward pass and errors at $\boldsymbol{\theta}^{(2)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$e_i = h_i - y_i$	$e_i\, x_{i1}$	$e_i\, x_{i2}$
1	3	0	1	$0.785 \to 0.8$	0.6900	0.3711	−0.3100	−0.9300	0
2	1	2	0	$-0.585 \to -0.6$	0.3543	0.4375	0.3543	0.3543	0.7086
3	3	1	1	$0.287 \to 0.3$	0.5744	0.5544	−0.4256	−1.2768	−0.4256
4	0	1	0	$-0.274 \to -0.3$	0.4256	0.5544	0.4256	0	0.4256
5	2	2	0	$-0.398 \to -0.4$	0.4013	0.5130	0.4013	0.8026	0.8026
				Column sums		2.4304	$S_0 = 0.4456$	$S_1 = -1.0499$	$S_2 = 1.5112$

(b) **[2 points]** Complete Table 8 with step 7, then finish step 8.

Table 8. Gradient and update, $\eta = 1.0$.

j	$\theta_j^{(2)}$	S_j (Table 7)	$\frac{\partial J}{\partial \theta_j} = \frac{S_j}{5}$	$\theta_j^{(3)} = \theta_j^{(2)} - \eta \frac{\partial J}{\partial \theta_j}$
0	0.224	0.4456	0.0891	0.135
1	0.187	−1.0499	−0.2100	0.397
2	−0.498	1.5112	0.3022	−0.800

$J(\boldsymbol{\theta}^{(2)}) = \frac{1}{5}\sum_i \ell_i =$ 2.4304/5 = 0.4861 Fans with $\hat{y}_i = y_i$ under $\boldsymbol{\theta}^{(2)}$: 5 of 5

Solution

$\boldsymbol{\theta}^{(2)}$ classifies the five fans correctly, with $J = 0.4861$. The gradient has shrunk. $|S_2|$ was 2.8158 in iteration 1 and is 1.5112 here. Steps get smaller as the fit improves.

Part 4: The Trained Model (5 points)

(a) **[2 points]** Evaluate $\boldsymbol{\theta}^{(3)}$ on the training data with the first three steps, then compute $J(\boldsymbol{\theta}^{(3)})$ and fill the prediction columns.

Table 9. Training data under $\boldsymbol{\theta}^{(3)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$\hat{y}_i$	Correct?
1	3	0	1	$1.326 \to 1.3$	0.7858	0.2410	1	yes
2	1	2	0	$-1.068 \to -1.1$	0.2497	0.2873	0	yes
3	3	1	1	$0.526 \to 0.5$	0.6225	0.4741	1	yes
4	0	1	0	$-0.665 \to -0.7$	0.3318	0.4032	0	yes
5	2	2	0	$-0.671 \to -0.7$	0.3318	0.4032	0	yes
					Column sum	1.8088		

$$J(\boldsymbol{\theta}^{(3)}) = \tfrac{1}{5}\textstyle\sum_i \ell_i = \quad 1.8088/5 = 0.3618$$

(b) **[1 point]** Copy your results into Table 10.

Table 10. Training history.

t	$\theta_0^{(t)}$	$\theta_1^{(t)}$	$\theta_2^{(t)}$	$J(\boldsymbol{\theta}^{(t)})$	Fans correct (of 5)
0	0.5	-0.5	0.5	1.1776	0
1	0.354	-0.083	-0.063	0.7361	2
2	0.224	0.187	-0.498	0.4861	5
3	0.135	0.397	-0.800	0.3618	5

(c) **[1 point]** A new fan reads $\mathbf{x}_{\text{new}} = (x_1, x_2) = (2, 1)$. Apply $\boldsymbol{\theta}^{(3)}$.

$$z_{\text{new}} = \theta_0 + 2\theta_1 + \theta_2 = \quad 0.135 + 0.794 - 0.800 = 0.129$$

$$h_{\text{new}} = g(z_{\text{new}}) = \quad g(0.1) = 0.5250$$

Predicted class $\hat{y}_{\text{new}}$: 1 (failure)

(d) **[1 point]** The dashed line in Figure 1 is the boundary of $\boldsymbol{\theta}^{(0)}$, namely $x_2 = x_1 - 1$. Compute two points on the boundary of $\boldsymbol{\theta}^{(3)}$ with two decimals, then draw the line through them.

At $x_1 = 0$: $x_2 =$ 0.17 At $x_1 = 3$: $x_2 =$ 1.66

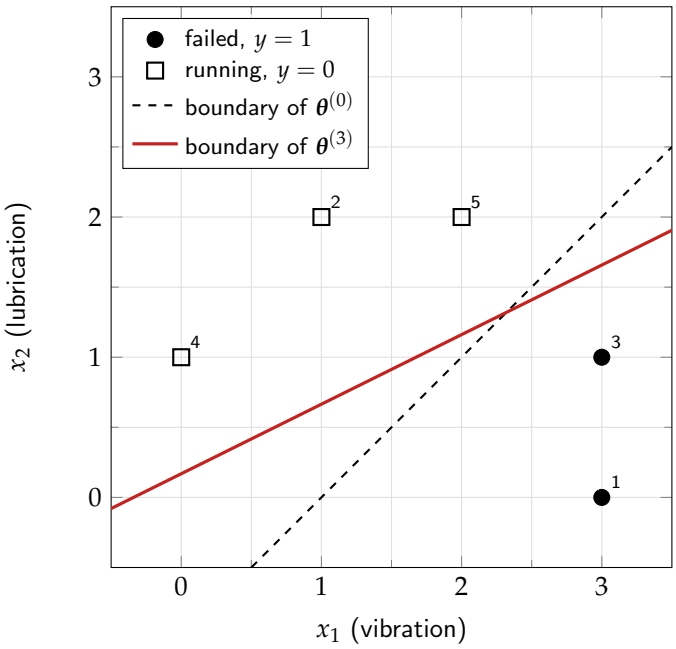


Figure 1. The five training fans, labeled by i. Draw the boundary of $\theta^{(3)}$ here.

Solution

(a) The scores are 1.326, -1.068, 0.526, -0.665 and -0.671, and their signs match the labels. $J(\theta^{(3)}) = 1.8088/5 = 0.3618$.

(b) During iteration 3 the count stayed at 5 of 5. J still dropped by 0.1243 and reached 0.3618. Cross-entropy charges for low confidence on a correct label, so J keeps falling after the labels come out right. These five points are linearly separable. Gradient descent run longer would keep growing $\|\theta\|$ and push J toward 0 without reaching it. A regularization term, or the stopping rule from the slides, keeps θ finite.

(c) $z_{\text{new}} = 0.135 + 0.794 - 0.800 = 0.129$, which rounds to 0.1, so $h_{\text{new}} = 0.5250$ and $\hat{y}_{\text{new}} = 1$. The model predicts failure with a probability barely above one half. I'd flag this fan for inspection. When a missed failure costs more than a false alarm, a threshold below 0.5 makes sense.

(d) $x_2 = (0.135 + 0.397\,x_1)/0.800$. At $x_1 = 0$, $x_2 = 0.16875 \approx 0.17$. At $x_1 = 3$, $x_2 = 1.326/0.800 = 1.6575 \approx 1.66$. The model predicts failure below the line. The dashed line of $\theta^{(0)}$ already separated the two classes. Its normal vector $(\theta_1, \theta_2) = (-0.5, 0.5)$ pointed toward the running fans, so the model called the healthy side "failure." The classifier depends on the sign of θ as well as on the line. Gradient descent cannot turn θ around in one step. It shrank θ toward the origin at $t = 1$ and came out with the opposite orientation.

Rounding check: exact arithmetic without the table gives $\theta^{(3)} = (0.1352,\ 0.4021,\ -0.8007)$ and $J(\theta^{(3)}) = 0.3637$. The table moves the answers in the third decimal.